

Tutorial: Semiconductors in Equilibrium

Fermi Level, Carrier Concentrations, Freeze-Out, and Compensation

Solid State Physics Lessons

October 26, 2025

Reference data (Si, unless stated)

- At $T = 300$ K: band gap $E_g \simeq 1.12$ eV, $n_i \simeq 1.0 \times 10^{10} \text{ cm}^{-3}$.
- Effective density of states: $N_C(300) \approx 2.8 \times 10^{19} \text{ cm}^{-3}$, $N_V(300) \approx 1.04 \times 10^{19} \text{ cm}^{-3}$.
- Temperature scaling (non-degenerate): $N_{C,V}(T) \propto T^{3/2}$, $kT/q \simeq 0.0259$ V at 300 K.
- Intrinsic level: $E_i = \frac{E_C + E_V}{2} + \frac{kT}{2} \ln(N_V/N_C)$.
- Mass-action law (equilibrium): $np = n_i^2$.
- Charge neutrality (full ionisation): $n + N_A = p + N_D$.
- Donor ionisation (with degeneracy $g_D \approx 2$):

$$N_D^+ = \frac{N_D}{1 + g_D e^{(E_F - E_D)/kT}}, \quad n = N_C e^{-(E_C - E_F)/kT}.$$

1 Part A Numerical Problems (10) with Enhanced Solutions

Problem 1: Intrinsic Fermi level position vs midgap

Problem Statement. Silicon at 300 K has $N_C = 2.8 \times 10^{19} \text{ cm}^{-3}$, $N_V = 1.04 \times 10^{19} \text{ cm}^{-3}$. Find the offset of the intrinsic Fermi level E_i from the *midgap* position $\frac{E_C + E_V}{2}$.

Solution.

Step 1: Understand the definition of intrinsic Fermi level.

The intrinsic Fermi level is the Fermi energy at which the electron and hole concentrations are equal ($n = p$). In equilibrium intrinsic material, this is also the true Fermi level. The exact formula includes a correction term from the density of states:

$$E_i = \frac{E_C + E_V}{2} + \frac{kT}{2} \ln \left(\frac{N_V}{N_C} \right).$$

The second term arises because the effective masses of electrons and holes differ, making $N_C \neq N_V$.

Step 2: Calculate the logarithmic term.

$$\ln\left(\frac{N_V}{N_C}\right) = \ln\left(\frac{1.04 \times 10^{19}}{2.8 \times 10^{19}}\right) = \ln(0.3714) \approx -0.990.$$

Step 3: Calculate the offset.

$$\Delta E = \frac{kT}{2} \ln\left(\frac{N_V}{N_C}\right) = \frac{0.0259}{2} \times (-0.990) = 0.01295 \times (-0.990) \approx -0.0128 \text{ eV} = -12.8 \text{ meV}.$$

Answer: E_i lies 12.8 meV below midgap (shifted toward the valence band).

Physical interpretation: - Midgap is exactly $(E_C + E_V)/2$ by definition - E_i is below midgap because $N_C > N_V$ (electron effective mass < hole effective mass) - This 12.8 meV shift is small but measurable and important for intrinsic material properties

Problem 2: n -type, fully ionised donors at 300 K

Problem Statement. n -type Si with $N_D = 1.0 \times 10^{16} \text{ cm}^{-3}$, $n_i = 1.0 \times 10^{10} \text{ cm}^{-3}$. Assuming complete ionisation, find the equilibrium carrier densities n_0 , p_0 and the Fermi level position $E_F - E_i$.

Solution.

Step 1: Apply charge neutrality.

In fully ionized material, every donor atom contributes one free electron:

$$n_0 = N_D^+ \approx N_D = 1.0 \times 10^{16} \text{ cm}^{-3}.$$

(We assume $p_0 \ll n_0$, so $p_0 + N_D^+ \approx N_D^+$.)

Step 2: Apply mass-action law.

The mass-action law holds at thermal equilibrium regardless of doping:

$$n_0 p_0 = n_i^2 \implies p_0 = \frac{n_i^2}{n_0} = \frac{(1.0 \times 10^{10})^2}{1.0 \times 10^{16}} = \frac{10^{20}}{10^{16}} = 1.0 \times 10^4 \text{ cm}^{-3}.$$

The minority carrier (hole) density is reduced by 12 orders of magnitude compared to intrinsic!

Step 3: Calculate Fermi level position relative to intrinsic level.

The Fermi level in n -type material lies above E_i (electrons are majority). The relationship is:

$$E_F - E_i = kT \ln\left(\frac{n_0}{n_i}\right) = 0.0259 \times \ln\left(\frac{10^{16}}{10^{10}}\right) = 0.0259 \times \ln(10^6).$$

$$\ln(10^6) = 6 \ln(10) \approx 6 \times 2.3026 = 13.816.$$

$$E_F - E_i = 0.0259 \times 13.816 = 0.357 \text{ eV}.$$

Answer: $n_0 = 10^{16} \text{ cm}^{-3}$, $p_0 = 10^4 \text{ cm}^{-3}$, $E_F - E_i \approx 0.357 \text{ eV}$.

Physical insights: - Donor ionisation is complete at 300 K ($kT \approx 26 \text{ meV} \gg E_d \approx 45 \text{ meV}$) - Fermi level lies 0.357 eV above midgap (in upper half of bandgap) - Minority carrier suppression ($p_0 = 10^{-6} \times n_i$) is dramatic

Problem 3: *p*-type, fully ionised acceptors at 300 K

Problem Statement. *p*-type Si with $N_A = 5.0 \times 10^{16} \text{ cm}^{-3}$, $n_i = 1.0 \times 10^{10} \text{ cm}^{-3}$. Find p_0 , n_0 , and $E_i - E_F$.

Solution.

Step 1: Apply charge neutrality.

With complete ionisation of acceptors:

$$p_0 = N_A^- \approx N_A = 5.0 \times 10^{16} \text{ cm}^{-3}.$$

Step 2: Apply mass-action law.

$$n_0 = \frac{n_i^2}{p_0} = \frac{10^{20}}{5.0 \times 10^{16}} = 2.0 \times 10^3 \text{ cm}^{-3}.$$

Minority electrons are suppressed by $\sim 10^7$ fold!

Step 3: Calculate Fermi level position.

In *p*-type material, the Fermi level lies below E_i (holes are majority). The distance is:

$$E_i - E_F = kT \ln \left(\frac{p_0}{n_i} \right) = 0.0259 \times \ln \left(\frac{5.0 \times 10^{16}}{10^{10}} \right).$$

$$\ln(5.0 \times 10^6) = \ln(5) + 6 \ln(10) \approx 1.609 + 13.816 = 15.425.$$

$$E_i - E_F = 0.0259 \times 15.425 = 0.400 \text{ eV}.$$

Answer: $p_0 = 5.0 \times 10^{16} \text{ cm}^{-3}$, $n_0 = 2.0 \times 10^3 \text{ cm}^{-3}$, $E_i - E_F \approx 0.400 \text{ eV}$.

Comparison to Problem 2: - *n*-type: Fermi level 0.357 eV above intrinsic level -
p-type: Fermi level 0.400 eV below intrinsic level (not symmetric due to effective mass difference!) - Both show strong minority carrier suppression

Problem 4: Lightly *n*-doped near intrinsic

Problem Statement. Very lightly doped *n*-Si with $N_D = 1.0 \times 10^{11} \text{ cm}^{-3}$, $n_i = 1.0 \times 10^{10} \text{ cm}^{-3}$. Note that $N_D \approx 10n_i$, so we cannot assume $N_D \gg n_i$. Find n_0 , p_0 , $E_F - E_i$ without approximations.

Solution.

Step 1: Set up the exact equations.

Charge neutrality: $n_0 = p_0 + N_D$ (assuming full ionisation). Mass-action: $p_0 = n_i^2/n_0$. Substituting the second into the first:

$$n_0 = \frac{n_i^2}{n_0} + N_D \implies n_0^2 - N_D n_0 - n_i^2 = 0.$$

Step 2: Solve the quadratic equation.

Using the quadratic formula (choosing the positive root):

$$n_0 = \frac{N_D + \sqrt{N_D^2 + 4n_i^2}}{2}.$$

Step 3: Calculate numerically.

$$N_D^2 = (10^{11})^2 = 10^{22}, \quad 4n_i^2 = 4 \times (10^{10})^2 = 4 \times 10^{20}.$$

$$\begin{aligned}
N_D^2 + 4n_i^2 &= 10^{22} + 4 \times 10^{20} = 1.04 \times 10^{22}. \\
\sqrt{1.04 \times 10^{22}} &\approx 1.0198 \times 10^{11}. \\
n_0 &= \frac{10^{11} + 1.0198 \times 10^{11}}{2} \approx 1.01 \times 10^{11} \text{ cm}^{-3}.
\end{aligned}$$

Step 4: Calculate minority carrier density.

$$p_0 = \frac{n_i^2}{n_0} = \frac{10^{20}}{1.01 \times 10^{11}} \approx 9.9 \times 10^8 \text{ cm}^{-3}.$$

Step 5: Calculate Fermi level position.

$$\begin{aligned}
E_F - E_i &= kT \ln \left(\frac{n_0}{n_i} \right) = 0.0259 \times \ln \left(\frac{1.01 \times 10^{11}}{10^{10}} \right). \\
&= 0.0259 \times \ln(10.1) \approx 0.0259 \times 2.313 = 0.060 \text{ eV}.
\end{aligned}$$

Answer: $n_0 \approx 1.01 \times 10^{11} \text{ cm}^{-3}$, $p_0 \approx 9.9 \times 10^8 \text{ cm}^{-3}$, $E_F - E_i \approx 0.060 \text{ eV}$.

Important insights: - Cannot use approximation $n_0 \approx N_D$ directly; must solve quadratic - Fermi level lies only 60 meV above intrinsic level (vs. 357 meV in Problem 2) - This material is between intrinsic and fully doped regimes - Useful for understanding transition from intrinsic to extrinsic conduction

Problem 5: Compensated semiconductor (full ionisation)

Problem Statement. Si with *both* donors and acceptors: $N_D = 1.0 \times 10^{16} \text{ cm}^{-3}$, $N_A = 3.0 \times 10^{15} \text{ cm}^{-3}$, $n_i = 1.0 \times 10^{10} \text{ cm}^{-3}$. Find n_0 , p_0 , $E_F - E_i$.

Solution.

Step 1: Understand compensation.

Compensation occurs when both donors and acceptors are present. They partially "cancel" each other: - Donors contribute electrons - Acceptors capture these electrons, becoming negatively charged - Net doping: $N_D - N_A = 1.0 \times 10^{16} - 3.0 \times 10^{15} = 7.0 \times 10^{15} \text{ cm}^{-3}$ (n -type)

Step 2: Apply charge neutrality.

$$n_0 + N_A^- = p_0 + N_D^+.$$

With full ionisation: $N_A^- \approx N_A$, $N_D^+ \approx N_D$. Thus:

$$n_0 - p_0 = N_D - N_A = 7.0 \times 10^{15} \text{ cm}^{-3}.$$

Step 3: Solve with mass-action.

Substitute $p_0 = n_i^2/n_0$ into $n_0 - p_0 = 7.0 \times 10^{15}$:

$$n_0 - \frac{n_i^2}{n_0} = 7.0 \times 10^{15} \implies n_0^2 - 7.0 \times 10^{15} n_0 - 10^{20} = 0.$$

Using the quadratic formula:

$$n_0 = \frac{7.0 \times 10^{15} + \sqrt{(7.0 \times 10^{15})^2 + 4 \times 10^{20}}}{2}.$$

The discriminant is $(7.0 \times 10^{15})^2 + 4 \times 10^{20} \approx 4.9 \times 10^{31}$ (the $4n_i^2$ term is negligible).

$$n_0 \approx \frac{7.0 \times 10^{15} + 7.0 \times 10^{15}}{2} = 7.0 \times 10^{15} \text{ cm}^{-3}.$$

Step 4: Calculate minority carriers.

$$p_0 = \frac{n_i^2}{n_0} = \frac{10^{20}}{7.0 \times 10^{15}} = 1.43 \times 10^4 \text{ cm}^{-3}.$$

Step 5: Calculate Fermi level position.

$$E_F - E_i = kT \ln \left(\frac{n_0}{n_i} \right) = 0.0259 \times \ln \left(\frac{7.0 \times 10^{15}}{10^{10}} \right) \\ = 0.0259 \times \ln(7.0 \times 10^5) = 0.0259 \times (1.946 + 11.513) = 0.0259 \times 13.459 = 0.349 \text{ eV}.$$

Answer: $n_0 \approx 7.0 \times 10^{15} \text{ cm}^{-3}$, $p_0 \approx 1.43 \times 10^4 \text{ cm}^{-3}$, $E_F - E_i \approx 0.349 \text{ eV}$.

Key insights on compensation: - Net carrier density reduced from N_D to $N_D - N_A$ (effective doping reduces) - Minority carrier density increases: $p_0 = n_i^2 / (N_D - N_A)$ is larger than in Problem 2 - Fermi level drops: $E_F - E_i = 0.349 \text{ eV}$ (vs. 0.357 eV uncompensated) - Compensation is used industrially to fine-tune doping and reduce leakage currents

Problem 6: Check for degeneracy at high donor density

Problem Statement. Heavily doped Si with $N_D = 5.0 \times 10^{18} \text{ cm}^{-3}$ (fully ionised). Given $N_C = 2.8 \times 10^{19} \text{ cm}^{-3}$, determine whether the non-degenerate (Boltzmann) approximation is valid.

Solution.

Step 1: Understand degeneracy criterion.

The non-degenerate approximation assumes Boltzmann statistics and requires:

$$E_C - E_F \gtrsim 3kT.$$

At 300 K: $3kT \approx 77.7 \text{ meV}$. If $E_C - E_F$ is smaller, we need Fermi-Dirac statistics (degenerate regime).

Step 2: Calculate $E_C - E_F$.

Using the non-degenerate formula (valid as first approximation):

$$n_0 = N_C \exp \left(-\frac{E_C - E_F}{kT} \right) \implies E_C - E_F = kT \ln \left(\frac{N_C}{n_0} \right).$$

With $n_0 \approx N_D = 5.0 \times 10^{18}$:

$$E_C - E_F = 0.0259 \times \ln \left(\frac{2.8 \times 10^{19}}{5.0 \times 10^{18}} \right) = 0.0259 \times \ln(5.6).$$

$$\ln(5.6) \approx 1.722 \implies E_C - E_F = 0.0259 \times 1.722 = 0.0446 \text{ eV} = 44.6 \text{ meV}.$$

Step 3: Compare to criterion.

$$E_C - E_F = 44.6 \text{ meV} < 3kT \approx 77.7 \text{ meV}.$$

The Fermi level is only 44.6 meV below the conduction band edge this is quite close!

Answer: $E_C - E_F \approx 44.6 \text{ meV}$ **borderline non-degenerate.**

Physical significance: - At this doping level, the semiconductor begins to exhibit degenerate behavior - Band-tailing and band-gap narrowing effects become important - Fermi-Dirac statistics are more accurate than Boltzmann - Further doping ($N_D > 10^{19} \text{ cm}^{-3}$) enters fully degenerate regime - This is relevant for metal-like doped regions in devices (e.g., highly doped contact layers)

Problem 7: Temperature variation (fully ionised, n_i given)

Problem Statement. n -type Si with $N_D = 1.0 \times 10^{16} \text{ cm}^{-3}$ (fully ionised at all temperatures). At $T = 350 \text{ K}$, use $n_i(350 \text{ K}) = 3.0 \times 10^{11} \text{ cm}^{-3}$ (measured value). Find n_0 , p_0 , and $E_F - E_i$.

Solution.

Step 1: Calculate thermal voltage at 350 K.

$$kT/q = \frac{350}{300} \times 0.0259 = 1.1667 \times 0.0259 = 0.0302 \text{ eV}.$$

Step 2: Apply full ionisation (valid since $kT > E_d$).

$$n_0 = N_D = 1.0 \times 10^{16} \text{ cm}^{-3}.$$

Step 3: Calculate minority carrier density.

At higher temperature, n_i increases dramatically (from 10^{10} at 300 K to 3×10^{11} at 350 K):

$$p_0 = \frac{n_i^2}{n_0} = \frac{(3.0 \times 10^{11})^2}{1.0 \times 10^{16}} = \frac{9.0 \times 10^{22}}{10^{16}} = 9.0 \times 10^6 \text{ cm}^{-3}.$$

Compare: $p_0 = 10^4 \text{ cm}^{-3}$ at 300 K $\rightarrow p_0 = 9 \times 10^6 \text{ cm}^{-3}$ at 350 K (increased by 900!)

Step 4: Calculate Fermi level position.

$$E_F - E_i = kT \ln \left(\frac{n_0}{n_i} \right) = 0.0302 \times \ln \left(\frac{10^{16}}{3.0 \times 10^{11}} \right).$$

$$= 0.0302 \times \ln(3.33 \times 10^4) = 0.0302 \times (1.204 + 9.210) = 0.0302 \times 10.414 = 0.314 \text{ eV}.$$

Compare: $E_F - E_i = 0.357 \text{ eV}$ at 300 K $\rightarrow E_F - E_i = 0.314 \text{ eV}$ at 350 K (decreased by 43 meV!)

Answer: $n_0 = 1.0 \times 10^{16} \text{ cm}^{-3}$, $p_0 = 9.0 \times 10^6 \text{ cm}^{-3}$, $E_F - E_i \approx 0.314 \text{ eV}$.

Temperature effects: - Majority carrier (n_0) unchanged (still equal to N_D , assuming ionisation remains complete) - Minority carrier (p_0) increases rapidly (proportional to n_i^2 , which grows exponentially with T) - Fermi level moves toward intrinsic level (E_i) as temperature increases - At very high T , all doped semiconductors approach intrinsic behavior

Problem 8: Freeze-out in n -Si (incomplete ionisation)

Problem Statement. n -type Si at low temperature ($T = 150 \text{ K}$) with $N_D = 1.0 \times 10^{16} \text{ cm}^{-3}$. Donor ionisation energy: $E_d = E_C - E_D = 45 \text{ meV}$, degeneracy factor $g_D = 2$. At 150 K: $kT = 12.9 \text{ meV}$ and $N_C(150) = 9.9 \times 10^{18} \text{ cm}^{-3}$. Find the ionisation fraction and carrier density (freeze-out regime).

Solution.

Step 1: Understand freeze-out physics.

At room temperature (300 K), donors are fully ionised because $kT = 26 \text{ meV} \gg E_d = 45 \text{ meV}$. At 150 K, $kT = 12.9 \text{ meV} \ll E_d = 45 \text{ meV}$. Most donors remain neutral (freeze-out regime).

The ratio $E_d/kT = 45/12.9 \approx 3.5$ determines the ionisation fraction.

Step 2: Set up the self-consistent equations.

Two key relationships:

$$n = N_C e^{-E_d/kT} \cdot \frac{1}{y}, \quad (1)$$

$$n = \frac{N_D}{1 + g_D y}, \quad (2)$$

where $y = e^{(E_F - E_D)/kT}$ is the parameter controlling ionisation.

The first equation comes from:

$$n = N_C e^{-(E_C - E_F)/kT} = N_C e^{-(E_C - E_D)/kT} e^{(E_D - E_F)/kT} = N_C e^{-E_d/kT} \cdot \frac{1}{y}.$$

The second is the Fermi-Dirac formula for ionised donors.

Step 3: Use the simpler freeze-out approximation.

In deep freeze-out, most donors are neutral ($N_D^+ \ll N_D$). The neutrality becomes:

$$n \approx N_D^+, \quad p \approx 0.$$

From the ionisation equation:

$$N_D^+(N_D - N_D^+) = g_D N_C e^{-E_d/kT} N_D^+.$$

Simplifying (assume $N_D^+ \ll N_D$, so $N_D - N_D^+ \approx N_D$):

$$(N_D - N_D^+) \approx N_D \implies n^2 \approx g_D N_C N_D e^{-E_d/kT}.$$

Taking the square root:

$$n \approx \sqrt{N_C N_D} \cdot \sqrt{g_D} \cdot e^{-E_d/(2kT)}.$$

For $g_D = 2$, the $\sqrt{g_D}$ factor is ≈ 1.41 .

Step 4: Calculate numerically.

$$\sqrt{N_C N_D} = \sqrt{9.9 \times 10^{18} \times 1.0 \times 10^{16}} = \sqrt{9.9 \times 10^{34}} \approx 9.95 \times 10^{17} \text{ cm}^{-3}.$$

$$E_d/(2kT) = 45/(2 \times 12.9) = 45/25.8 \approx 1.744.$$

$$e^{-1.744} \approx 0.175.$$

$$n \approx 9.95 \times 10^{17} \times 1.41 \times 0.175 \approx 2.46 \times 10^{17} \text{ cm}^{-3}.$$

Alternatively, using $n \approx \sqrt{N_C N_D} \cdot e^{-E_d/(2kT)}$ (neglecting $\sqrt{g_D}$ for simplicity):

$$n \approx 9.95 \times 10^{17} \times 0.175 \approx 1.74 \times 10^{17} \text{ cm}^{-3}.$$

Step 5: Calculate ionisation fraction.

$$\text{Ionisation fraction} = \frac{n}{N_D} = \frac{1.74 \times 10^{17}}{1.0 \times 10^{16}} = 0.0174 \approx \boxed{1.74\%}.$$

Answer: At 150 K, only $\boxed{\sim 1.7\%}$ of donors are ionised, and $\boxed{n \approx 1.74 \times 10^{17} \text{ cm}^{-3}}$.

Physical insights on freeze-out: - Despite $N_D = 10^{16} \text{ cm}^{-3}$ (heavily doped at room temperature), the carrier density drops to $\sim 10^{17} \text{ cm}^{-3}$ (still significant!) - Resistance increases dramatically at low T due to reduced mobility and lower n - Freeze-out is exploited in cryogenic detectors (reduced dark current) - The exponential dependence $e^{-E_d/(2kT)}$ explains the steep resistivity rise at low T

Problem 9: Compensated n -type (full ionisation), Fermi level

Problem Statement. Compensated Si with $N_D = 5.0 \times 10^{16} \text{ cm}^{-3}$, $N_A = 2.0 \times 10^{16} \text{ cm}^{-3}$, $n_i = 1.0 \times 10^{10} \text{ cm}^{-3}$. Find $E_F - E_i$ at 300 K.

Solution.

Step 1: Determine net doping.

$$N_D - N_A = 5.0 \times 10^{16} - 2.0 \times 10^{16} = 3.0 \times 10^{16} \text{ cm}^{-3}.$$

This is still n -type, but with reduced effective doping.

Step 2: Apply charge neutrality and mass-action.

$$n_0 - p_0 = N_D - N_A = 3.0 \times 10^{16}, \quad p_0 = \frac{n_i^2}{n_0}.$$

Substituting:

$$n_0 - \frac{10^{20}}{n_0} = 3.0 \times 10^{16} \implies n_0^2 - 3.0 \times 10^{16} n_0 - 10^{20} = 0.$$

Step 3: Solve the quadratic.

$$n_0 = \frac{3.0 \times 10^{16} + \sqrt{(3.0 \times 10^{16})^2 + 4 \times 10^{20}}}{2} \approx \frac{3.0 \times 10^{16} + 3.0 \times 10^{16}}{2} = 3.0 \times 10^{16} \text{ cm}^{-3}.$$

(The discriminant is dominated by the first term.)

Step 4: Calculate Fermi level.

$$E_F - E_i = kT \ln \left(\frac{n_0}{n_i} \right) = 0.0259 \times \ln \left(\frac{3.0 \times 10^{16}}{10^{10}} \right).$$

$$= 0.0259 \times \ln(3.0 \times 10^6) = 0.0259 \times (1.099 + 13.816) = 0.0259 \times 14.915 = 0.386 \text{ eV}.$$

Answer: $\boxed{E_F - E_i \approx 0.386 \text{ eV}}$.

Comparison table:

Problem	N_D	N_A	$E_F - E_i$
2: Pure n -type	10^{16}	0	0.357 eV
5: Compensated	10^{16}	3×10^{15}	0.349 eV
9: Compensated	5×10^{16}	2×10^{16}	0.386 eV

The Fermi level position depends on the net doping ($N_D - N_A$), following $E_F - E_i = kT \ln(n_0/n_i)$.

Problem 10: Extracting $E_C - E_F$ from Hall-measured n

Problem Statement. Hall measurements give $n_0 = 2.0 \times 10^{15} \text{ cm}^{-3}$ at 300 K. From this data, determine: (i) The distance of Fermi level from conduction band: $E_C - E_F$ (ii) The Fermi level position relative to intrinsic level: $E_F - E_i$

Given: $N_C = 2.8 \times 10^{19} \text{ cm}^{-3}$, $n_i = 1.0 \times 10^{10} \text{ cm}^{-3}$.

Solution.

Step 1: Use the non-degenerate conduction band formula.

The carrier density from the conduction band is:

$$n = N_C \exp\left(-\frac{E_C - E_F}{kT}\right).$$

Rearranging:

$$E_C - E_F = kT \ln\left(\frac{N_C}{n}\right).$$

Step 2: Calculate $E_C - E_F$.

$$E_C - E_F = 0.0259 \times \ln\left(\frac{2.8 \times 10^{19}}{2.0 \times 10^{15}}\right) = 0.0259 \times \ln(1.4 \times 10^4).$$

$$\ln(1.4 \times 10^4) = \ln(1.4) + 4 \ln(10) \approx 0.336 + 9.210 = 9.546.$$

$$E_C - E_F = 0.0259 \times 9.546 = \boxed{0.247 \text{ eV}}.$$

The Fermi level lies 247 meV below the conduction band edge.

Step 3: Calculate $E_F - E_i$ using the definition.

$$E_F - E_i = kT \ln\left(\frac{n}{n_i}\right) = 0.0259 \times \ln\left(\frac{2.0 \times 10^{15}}{1.0 \times 10^{10}}\right).$$

$$= 0.0259 \times \ln(2.0 \times 10^5) = 0.0259 \times (0.693 + 11.513) = 0.0259 \times 12.206 = \boxed{0.316 \text{ eV}}.$$

The Fermi level lies 316 meV above the intrinsic level.

Answer: $E_C - E_F = 0.247 \text{ eV}$, $E_F - E_i = 0.316 \text{ eV}$.

Check: Internal consistency.

Verify: $E_C - E_F + E_F - E_i$ should relate to $E_C - E_i = E_g/2 + \text{offset}$:

$$0.247 + 0.316 = 0.563 \text{ eV}.$$

Comparing to $E_g/2 = 1.12/2 = 0.56 \text{ eV}$ (plus small offset from $N_C \neq N_V$), this is consistent.

Experimental significance: - Hall measurements give n_0 directly - From n_0 , we can extract the Fermi level position without knowing the doping concentration! - This is useful for characterising unknown samples - Combined with conductivity and mobility, we can determine material properties

2 Part B Conceptual Q&A (10 Questions)

Q1. Where is the intrinsic Fermi level in Si at 300 K relative to midgap?

Answer. Slightly below midgap by $\frac{kT}{2} \ln(N_V/N_C) \approx -13 \text{ meV}$ (shifted toward valence band).

Reason: $N_C > N_V$ due to smaller electron effective mass. The intrinsic level depends on the density of states, not just the bandgap.

Q2. What equations determine n_0 and p_0 in equilibrium?

Answer. Two fundamental equations:

1. **Mass-action law:** $n_0 p_0 = n_i^2$ (always holds at thermal equilibrium)
2. **Charge neutrality:** $n_0 + N_A^- = p_0 + N_D^+$ (conservation of charge)

With complete ionisation: $N_A^- \approx N_A$, $N_D^+ \approx N_D$. In the freeze-out regime, use $N_D^+ = N_D / (1 + g_D e^{(E_F - E_D)/kT})$.

Q3. How do you compute $E_F - E_i$ in non-degenerate n -type material?

Answer. Use the logarithmic relation:

$$E_F - E_i = kT \ln \left(\frac{n_0}{n_i} \right).$$

Similarly, for p -type: $E_i - E_F = kT \ln(p_0/n_i)$.

These follow directly from the definitions of the quasi-Fermi level and the mass-action law.

Q4. What is "freeze-out"?

Answer. The low-temperature regime where dopant ionisation becomes incomplete because $kT \ll E_d$ (ionisation energy).

- Carrier densities drop approximately as $n \sim \sqrt{N_C N_D} e^{-E_d/(2kT)}$
- Material transitions from extrinsic (doping-controlled) to intrinsic conduction at very low T
- Resistivity rises steeply with decreasing T in the freeze-out regime
- Used in cryogenic detectors to minimize dark currents

Q5. What is a "compensated" semiconductor?

Answer. A crystal with both donors and acceptors present. The net doping is $N_D - N_A$ (n -type if positive, p -type if negative).

- Effective doping reduces: $n_0 - p_0 = N_D - N_A$ (not N_D alone)
- Minority carrier density *increases*: $p_0 = n_i^2/(N_D - N_A) > n_i^2/N_D$
- Used to fine-tune resistivity and suppress leakage currents
- Common in CMOS technology (low-doped regions are compensated)

Q6. When does the non-degenerate (Boltzmann) approximation break down?

Answer. When $E_C - E_F \lesssim 3kT$ for electrons (or $E_F - E_V \lesssim 3kT$ for holes).

- At 300 K, this criterion is $E_C - E_F < 78$ meV (approximately)
- At very high doping ($N_D > 10^{18}$ cm⁻³ in Si), the Fermi level enters the conduction band edge
- Fermi-Dirac statistics become necessary; band-gap narrowing and band-tailing effects appear
- Relevant for ohmic contacts and highly doped regions

Q7. How do effective masses affect the Fermi level in intrinsic Si?

Answer. Through the effective density of states:

$$N_C \propto m_e^{*3/2}, \quad N_V \propto m_h^{*3/2}.$$

Since $m_e^* < m_h^*$ in Si, we have $N_C > N_V$, so E_i is shifted *below* midgap toward the valence band by $\Delta E = \frac{kT}{2} \ln(N_V/N_C) \approx -13$ meV.

This is a fundamental asymmetry in intrinsic semiconductors with unequal masses.

Q8. Does $np = n_i^2$ hold in compensated materials?

Answer. Yes. The mass-action law holds at thermal equilibrium *regardless of compensation*. Compensation affects:

- The neutrality equation (how n , p , N_D , N_A are related), not the mass-action law
- The net doping: $n - p = N_D - N_A$ (not N_D)
- The Fermi level position: $E_F - E_i = kT \ln(n/n_i) = kT \ln[(N_D - N_A)/n_i]$ (in extrinsic regime)

Compensation reduces minority carrier suppression: $p_0 = n_i^2/(N_D - N_A)$ is larger.

Q9. How does temperature affect n_i and the Fermi level in extrinsic n -Si?

Answer.

- n_i grows **very rapidly** with T (exponential dependence): $n_i \propto T^{3/2} e^{-E_g/(2kT)}$
- In extrinsic n -type: $n_0 \approx N_D$ remains constant (if ionisation complete)
- Minority holes: $p_0 = n_i^2/N_D$ **increases** dramatically
- Fermi level: $E_F - E_i = kT \ln(N_D/n_i)$ **decreases** with T (moves toward E_i)
- At very high T , material transitions to intrinsic: $n \approx p \approx n_i$

Q10. How to include incomplete ionisation in the neutrality equation?

Answer. Replace constant doping concentrations with ionisation-dependent quantities:

$$n + N_A^- = p + N_D^+,$$

where:

$$N_D^+ = \frac{N_D}{1 + g_D e^{(E_F - E_D)/kT}}, \quad N_A^- = \frac{N_A}{1 + g_A e^{(E_A - E_F)/kT}}.$$

Then solve simultaneously with:

$$n = N_C e^{-(E_C - E_F)/kT}, \quad p = N_V e^{-(E_F - E_V)/kT}, \quad np = n_i^2.$$

This is a system of transcendental equations typically solved iteratively. The freeze-out regime has $N_D^+ \ll N_D$, simplifying to analytic approximations like $n \approx \sqrt{N_C N_D} e^{-E_d/(2kT)}$.